

Lecture 8

Rate of Convergence: An iterative method is said to be of order p or the rate of convergence is p , if p is the largest positive real number for which there exists a finite constant $C \neq 0$ such that

$$|\epsilon_{k+1}| \leq C|\epsilon_k|^p,$$

Rate of Convergence of Newton-Raphson method

Newton-Raphson iteration formula

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \quad k = 0, 1, 2, \dots$$

On substituting the value $x_k = \xi + \epsilon_k$ and expanding the functions $f(\xi + \epsilon_k)$ and $f'(\xi + \epsilon_k)$, by Taylor' series about the point ξ and neglecting the 3rd and higher order error terms, we get

$$\xi + \epsilon_{k+1} = (\xi + \epsilon_k) - \frac{f(\xi) + \epsilon_k f'(\xi) + \frac{1}{2} \epsilon_k^2 f''(\xi)}{f'(\xi) + \epsilon_k f''(\xi) + \frac{1}{2} \epsilon_k^2 f'''(\xi)}$$

As ξ is the exact root of $f(x) = 0$, therefore $f(\xi) = 0$ gives

$$\epsilon_{k+1} = \epsilon_k - \frac{\epsilon_k f'(\xi) + \frac{1}{2} \epsilon_k^2 f''(\xi)}{f'(\xi) + \epsilon_k f''(\xi) + \frac{1}{2} \epsilon_k^2 f'''(\xi)}$$

So

$$\epsilon_{k+1} = \epsilon_k - \frac{\epsilon_k + \frac{1}{2} \epsilon_k^2 \frac{f''(\xi)}{f'(\xi)}}{1 + \epsilon_k \frac{f''(\xi)}{f'(\xi)} + \frac{1}{2} \epsilon_k^2 \frac{f'''(\xi)}{f'(\xi)}}$$

$$\epsilon_{k+1} = \epsilon_k - \left[\epsilon_k + \frac{1}{2} \epsilon_k^2 \frac{f''(\xi)}{f'(\xi)} \right] \left[1 + \left(\epsilon_k \frac{f''(\xi)}{f'(\xi)} + \frac{1}{2} \epsilon_k^2 \frac{f'''(\xi)}{f'(\xi)} \right) \right]^{-1}$$

Expanding the binomial expansion, we get

$$\epsilon_{k+1} = \frac{1}{2} \epsilon_k^2 \frac{f''(\xi)}{f'(\xi)} + O(\epsilon_k^3)$$

As ϵ_k is sufficiently small so neglecting ϵ_k^3 and the higher power terms, we get

$$\epsilon_{k+1} = C \epsilon_k^2,$$

where $C = \frac{1}{2} \frac{f''(\xi)}{f'(\xi)}$.

Thus on comparing with the equation $\epsilon_{k+1} = C \epsilon_k^p$, we find $p = 2$, therefore Newton-Raphson method has second order of convergence.

Numerical Problem

Example: The iterative formula for the solution of equation $x = \sqrt{a}$, is given as follows:

$$x_{k+1} = \frac{1}{2}x_k \left(1 + \frac{a}{x_k^2} \right),$$

show that the rate of convergence for this iterative method is 2.

Solution: Let the actual root be ξ of given equation, therefore $\xi^2 = a$. Assume that the error in the k-th approximation is ϵ_k , therefore substituting $x_k = \xi + \epsilon_k$ and $x_{k+1} = \xi + \epsilon_{k+1}$ in the given iterative formula, we get

Cont...

$$\begin{aligned}\xi + \epsilon_{k+1} &= \frac{1}{2}(\xi + \epsilon_k) \left(1 + \frac{\xi^2}{(\xi + \epsilon_k)^2} \right) \\ &= \frac{1}{2}(\xi + \epsilon_k) \left[1 + \left(1 + \frac{\epsilon_k}{\xi} \right)^{-2} \right] \\ &= \frac{1}{2}(\xi + \epsilon_k) \left[2 - \frac{2\epsilon_k}{\xi} + \frac{3\epsilon_k^2}{\xi^2} - \dots \right] \\ &= \frac{1}{2} \left[2\xi + (2 - 2)\epsilon_k + (3 - 2)\frac{\epsilon_k^2}{\xi} + \dots \right]\end{aligned}$$

Therefore,

$$\epsilon_{k+1} = \frac{1}{2\xi}\epsilon_k^2 + O(\epsilon_k^3).$$

Hence the method has second order of convergence with error constant as $1/2\xi$.